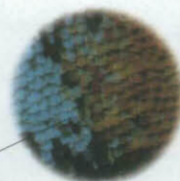


BUTTERFLIES AND MOTHS



SCALY WINGS AND A COILED feeding tube set butterflies and moths apart from other insects. Together, they form a single group of about 170,000 species, of which 90 per cent are moths. Both have four stages to their life cycle in which they change from a caterpillar to an adult with wings. They feed on plants, and rely on camouflage, irritating hairs or spines, or poisons in their body for protection against predators.

Scales overlap like the tiles on a roof.



Wing scales
Scales on the wings contain coloured pigments. Some scales produce colours by reflecting the light.



The front and back wings of a moth are hooked together.

Moth's bright colouring indicates it is poisonous

Moths

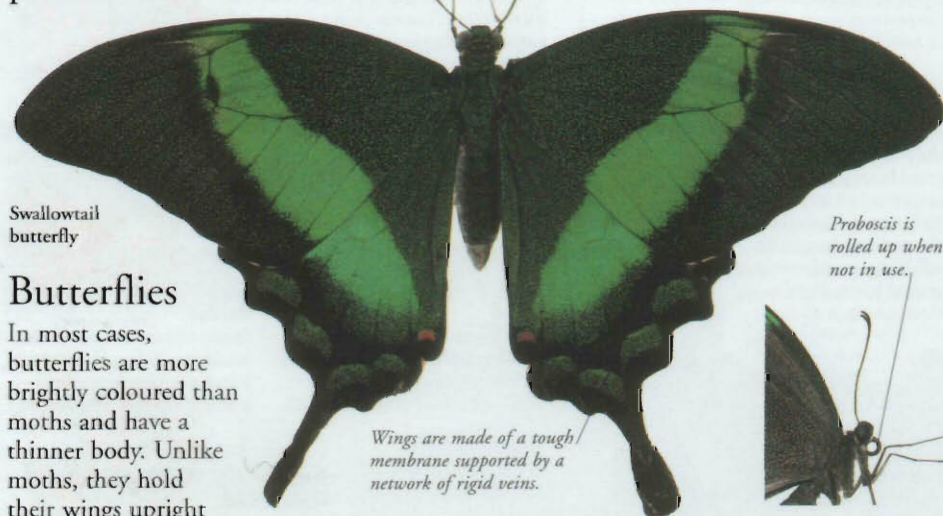
Zygaenid moth

Most moths fly at night. They tend to have drab colours, and have a fatter body and longer, narrower wings than butterflies. When resting, moths usually hold their wings open or fold them flat over their back.

Swallowtail butterfly

Butterflies

In most cases, butterflies are more brightly coloured than moths and have a thinner body. Unlike moths, they hold their wings upright when resting. The front and back wings are loosely joined together by a lobe on the back wing that grips the front wing. Butterflies are usually active by day rather than by night.



Wings are made of a tough membrane supported by a network of rigid veins.

Proboscis is rolled up when not in use.



Moth antennae have a large surface area for picking up scents.



Feeding tube

Adult butterflies and moths suck up liquid food, such as flower nectar, through a tube called a proboscis. A few moths have no proboscis because they do not feed as adults.

Antennae

Insects use their antennae for smelling, touching, and tasting. Butterfly antennae are clubbed; moth antennae range from single strands to feathery branches.

Life cycle

Butterflies and moths start life as an egg, which hatches into a caterpillar. This feeds and grows until it turns into a pupa. The adult develops inside the pupa. This process of change is called metamorphosis.



Pupa protects developing adult.

Adult butterfly emerges.

Henry Bates

Henry Walter Bates (1825–92) was a British naturalist and explorer who studied camouflage in animals. He found that some harmless insects

look the same as a poisonous insect so that predators leave them alone. This is now called Batesian mimicry, after Henry Bates.



Defence

To escape from predators, butterflies and moths often fly away or hide. Some have irritating hairs or spines, or are poisonous. Bright colours may warn predators that a butterfly or moth is poisonous. Poisons often build up in a caterpillar from the plant it eats. These then remain in the adult.



Camouflage

Many butterflies and moths blend in with their surroundings at some stage of their life cycle. Camouflaged like this, they may escape predators.

Eyespots

False eyes on the wings can startle predators or stop them from pecking the real eyes. A damaged wing is not as serious as an injury to the head.

Mimicry

Some butterflies and moths gain protection by looking like another species of butterfly or moth. The top butterfly shown here is poisonous; the bottom one is not.



Butterfly pumps blood into its wings to expand and stiffen them.

Adult Blue Morpho



SWALLOWTAIL BUTTERFLY

SCIENTIFIC NAME *Papilio palinurus*

ORDER Lepidoptera

FAMILY Papilionidae

DISTRIBUTION From Burma to the islands of Borneo and the Philippines in Southeast Asia

HABITAT Tropical rainforest

DIET Flower nectar

SIZE Wing span: 9.5 cm (3.75 in)

LIFESPAN Varies (The adults of most butterflies live for only a few weeks or months)

Wing colour

When a butterfly is resting, only the underside of its wings shows. This is often coloured for camouflage. The colours of the upper side help to attract a mate.



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